

Rohan Bajpai

U.S. Citizen | 571-442-7276 | rohanibajpai@gmail.com | [linkedin.com/in/rohan-bajpai-20](https://www.linkedin.com/in/rohan-bajpai-20) | github.com/rohannbajpai

EDUCATION

University of Maryland

College Park, MD

Bachelor of Science in Computer Science

August 2021 – December 2024

- **GPA:** 3.7
- **Honors:** CS Departmental Honors, Presidential Scholarship Recipient
- **Relevant Coursework:** Introduction to Machine Learning, Introduction to Parallel Computing, Database Design, Computer Networks, Algorithms, Compilers, Applied Linear Algebra, Applied Probability & Statistics
- **Clubs and Organizations:** Software Engineer at Hack4Impact, Webmaster at Phi Chi Theta

TECHNICAL SKILLS

Languages: Kotlin, Java, Python, C/C++, HTML, JavaScript, SASS, CSS, SQL

Technologies: Flask, Spring Boot, Node.js, Django, Elastic, Gym, PySpark, Pandas, MongoDB, Postgres, MySQL, React, Kafka

Tools: Git, Docker, GitLab, GitHub, AWS (Certified Developer)

EXPERIENCE

Amazon Web Services

Arlington, VA

Software Development Engineer Intern

May 2023 – August 2023, May 2024 – August 2024

- Developed a classification model using AWS SageMaker, Spark, and Python to identify a **2.62%** error rate in tier-1 infrastructure data across over **1,200,000** servers, enhancing cross-service reliability within AWS.
- Automated MySQL credential rotation via API, eliminating a **10+** day manual process, improving security and efficiency.
- Implemented a Single Sign-On (SSO) system, reducing customer beta testing onboarding time by **90%**.
- Collaborated with cross-functional teams, enhancing communication and delivering solutions ahead of schedule.
- Received organizational award for outstanding performance during 12-week internship.

Capital One

College Park, MD

Software Engineer Intern - Machine Learning

September 2023 – April 2024

- Architected and deployed a distributed real-time classification model using AWS Step Functions and DynamoDB, predicting account-customer mappings with sub-second latency and **99.99%** uptime.
- Generated over **2 million** synthetic data points via statistical sampling using PySpark and reverse feature engineering for performance testing, ensuring system robustness.
- Collaborated closely with data science and engineering teams to implement algorithms and refine system architecture.

Leidos

Reston, VA

Software Engineering Co-op

June 2022 – May 2023

- Developed a full-stack web application and API for predictive analytics visualization using Spring Boot and PostgreSQL, enhancing data accessibility for clients.
- Implemented single-sign-on with Elasticsearch/Kibana, improving user experience by **50%** based on customer feedback.
- Applied AI/ML research to design a reinforcement learning-based automated firewall hardening product using Python and Gym, improving network security measures.

PROJECTS

Plato | Python, AWS, LangChain, Pinecone

December 2023 – Present

- Launched "Plato," enabling engineers to seamlessly search, understand, and integrate internal microservices
- Interviewed 20+ potential customers to initially identify pain-points and areas for improvement
- Implemented novel Retrieval-Augmented Generation (RAG) and re-ranking techniques
- Invited to interview with YCombinator, Summer 2023

Melp | Python, Flask, React Native, Node.js, MongoDB

Spring 2022 – December 2022

- Developed "Melp," a mobile app for roommates, featuring bill-splitting, location-sharing, messaging, and chore tracking
- Created a Flask-based service using Tesseract for receipt OCR and OpenCV for image preprocessing